

Survival Analysis Report: Telco Churn and Customer Lifetime Value

1. Model Summary and Interpretation

We applied three parametric Accelerated Failure Time (AFT) models—**Weibull**, **LogNormal**, and **LogLogistic**—to analyze churn duration in a telco customer dataset. Based on model comparison metrics:

- **LogNormal AFT** was selected as the final model due to its lowest AIC (2963.64), highest log-likelihood, and interpretability.
- The model achieved a **concordance index of 0.78**, indicating good predictive ability in estimating time-to-churn.

Key statistically significant features ($p < 0.05$) influencing customer survival time:

- **Increase survival (positive impact):**
 - address: Longer residence duration
 - age: Older customers
 - custcat: Higher-tier customer categories
- **Decrease survival (increase churn risk):**
 - internet: Having internet service
 - voice: Having voice service
 - marital: Being unmarried
 - forward: Using call forwarding

These findings suggest that customers who are older and more stable in their residence, and those in premium service categories, tend to stay longer. Conversely, subscribers with active services (internet/voice), or who are unmarried, churn sooner.

2. Most Valuable Customer Segments

Customer Lifetime Value (CLV) was calculated as:

$$\text{CLV} = \text{Predicted Lifetime} \times \text{Income} \times 1,000$$

CLV distribution was highly skewed, with some customers exhibiting extremely high lifetime value. We identified the **most valuable segments** as:

- Customers with **long predicted tenure and high income**
- Customers **without internet/voice services**, who have lower churn risk
- Subscribers in **premium customer categories**

These customers represent the top quartile of CLV and should be the primary focus for customer retention and upselling strategies.

3. Retention Strategy and Budget Recommendation

Using the LogNormal model, we predicted each customer's **12-month survival probability**. Customers with less than 50% probability of staying were flagged as "at risk."

- Total number of **at-risk customers**: *[Insert number from output]*
- Total CLV of at-risk group: *[Insert total CLV value from output]*
- Suggested annual retention budget (assuming 5% of CLV):

$$\text{Budget} \approx 5\% \times \text{At-Risk CLV} = [\text{Insert amount}]$$

This approach ensures budget is targeted toward high-value customers at the highest churn risk.

4. Strategic Recommendations

To reduce churn and increase long-term revenue, we recommend the following:

- **Targeted retention offers** for customers with internet/voice bundles and high churn probabilities
- **Loyalty incentives** for unmarried or recently joined customers
- **Proactive engagement** with high-CLV customers before they reach critical churn windows
- **Continuous monitoring** of survival probabilities and recalibration of the model quarterly